

Kinematical Equations

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Kinematic Vector Definitions

Each of the kinematic quantities (position, displacement, velocity, etc.) are defined as follows:

Position Vector

The position vector at time t is given by:

$$\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

Displacement Vector

The displacement vector from time t_1 to time t_2 is:

$$\begin{aligned} \vec{s}(t_1, t_2) &= \vec{r}(t_2) - \vec{r}(t_1) \\ &= \begin{bmatrix} x(t_2) - x(t_1) \\ y(t_2) - y(t_1) \end{bmatrix} \end{aligned}$$

Velocity Vector

The instantaneous velocity at time t is:

$$\begin{aligned} \vec{v}(t) &= \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} \\ &= \frac{d\vec{r}}{dt} \end{aligned}$$

Acceleration Vector

The instantaneous acceleration at time t is:

$$\begin{aligned} \vec{a}(t) &= \lim_{\Delta t \rightarrow 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t} \\ &= \frac{d\vec{v}}{dt} \end{aligned}$$

Component-wise Representation

The position vector can be written in component form as:

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}$$

By differentiating with respect to time:

$$\begin{aligned} \vec{v}(t) &= \frac{dx(t)}{dt}\hat{i} + x(t)\frac{d\hat{i}}{dt} + \frac{dy(t)}{dt}\hat{j} + y(t)\frac{d\hat{j}}{dt} \\ &= \frac{dx(t)}{dt}\hat{i} + \frac{dy(t)}{dt}\hat{j} \end{aligned}$$

Note that $\frac{d\hat{i}}{dt} = \frac{d\hat{j}}{dt} = 0$ since the Cartesian basis vectors are constant (independent of time).

The acceleration can similarly be decomposed:

$$\vec{a}(t) = \frac{d^2x(t)}{dt^2}\hat{i} + \frac{d^2y(t)}{dt^2}\hat{j}$$

These equations can also be extended to three dimensions by adding a z -component with basis vector $\hat{k}$.

Formal Solution of Kinematic Equations

Suppose $\vec{a}(t)$ is known. How do we determine $\vec{r}(t)$ and $\vec{v}(t)$?

In three dimensions, the acceleration vector is:

$$\begin{aligned}\vec{a}(t) &= \frac{dv_x(t)}{dt} \hat{i} + \frac{dv_y(t)}{dt} \hat{j} + \frac{dv_z(t)}{dt} \hat{k} \\ &= a_x(t) \hat{i} + a_y(t) \hat{j} + a_z(t) \hat{k}\end{aligned}$$

Focusing on the x -component:

$$\begin{aligned}\frac{dv_x(t)}{dt} &= a_x(t) \\ \int_{v_x(t_0)}^{v_x(t)} dv_x &= \int_{t_0}^t a_x(t') dt' \\ v_x(t) - v_x(t_0) &= \int_{t_0}^t a_x(t') dt' \\ v_x(t) &= v_x(t_0) + \int_{t_0}^t a_x(t') dt'\end{aligned}$$

Returning to three dimensions, the velocity vector is:

$$\vec{v}(t) = \vec{v}(t_0) + \int_{t_0}^t \vec{a}(t') dt'$$

Similarly, we can determine the position vector. Starting from $\frac{d\vec{r}}{dt} = \vec{v}(t)$:

$$\vec{r}(t) = \vec{r}(t_0) + \int_{t_0}^t \vec{v}(t') dt'$$

These integral equations allow us to find the velocity and position given the acceleration and appropriate initial conditions $\vec{v}(t_0)$ and $\vec{r}(t_0)$.